Ideal and Non-ideal solution

1. (b) In solution showing positive type of deviation the partial pressure of each component of solution is greater than the vapour pressure as expected according to Raoult's law.

In solution of methanol & benzene methanol molecules are held together due to hydrogen bonding as shown below.

$$\begin{array}{c} CH_3CH_3CH_3 \\ ||| \\ ----O-H----O-H----O-H---- \end{array}$$

On adding benzene, the benzene molecules get in between the molecule of methanol thus breaking the hydrogen bonds. As the resulting solution has weak intermolecular attraction, the escaping tendency of alcohol & benzene molecule from the solution increases. Consequently the vapour pressure of the solution is greater than the vapour pressure as expected from Raoult's law.

2. **(d)** $CCl_4 + CHCl_3$

Explanation:

A non-ideal solution is formed when intermolecular forces between unlike molecules are different from those between like molecules.

- Benzene + toluene → nearly ideal
- n-hexane + n-heptane → ideal
- Ethyl bromide + ethyl iodide → nearly ideal
- $CCl_4 + CHCl_3$→ non-ideal due to specific intermolecular interactions

Hence, $CCl_4 + CHCl_3$forms a non-ideal solution.

3. (b) Chloroform & acetone form a non-ideal solution, in which $A.....B$ type interaction are more than $A......A$ & $B......B$ type interaction due to H -bonding. Hence, the solution shows, negative deviation from Raoult's Law *i.e.,*

$\Delta V_{mix} = -ve;$ $\Delta H_{mix} = -ve$

∴total volume of solution = less than (30 + 50 *ml*)

or $< 80ml$

4. (b) H_2O and C_4H_9OH do not form ideal solution because there is hydrogen bonding between H_2O and C_4H_9OH.

6. (a) Aromatic compound generally separated by fractional distillation. *e.g.* Benzene + Toluene.

7. (d) C_2H_5I and C_2H_5OH do not form ideal solution.

8 **(d) All of these**

Explanation:

ideal solution:

- obeys Raoult's law
- has no heat change on mixing
- has no volume change on mixing

9 **(c) Negative deviation with non-ideal solution**

Explanation:

Negative deviation lowers vapour pressure, so boiling point increases.

10 **(d) A–B interaction stronger than A–A and B–B interaction**

Explanation:

Stronger intermolecular attraction decreases escaping tendency, causing negative deviation.

11 **(a) 1 M NaCl**

Explanation:

NaCl dissociates into ions, so solution becomes non-ideal.

12 **: (c) Ideal**

13 **(a) n-heptane and n-hexane**

Explanation:

Chemically similar liquids generally form ideal solutions.

14 **(a)** $(CH_3)_2CO+C_2H_5OH$

Explanation:

Positive deviation occurs when A–B interactions are weaker.

15 **(d)** C_2H_5I **and** C_2H_5OH

Explanation:

Ethanol forms hydrogen bonding, causing non-ideal behaviour.

16 **(c)**

Explanation:

For solution formation:

$$\Delta H_{soln} = \Delta H_1 + \Delta H_2 - \Delta H_3$$

For ideal solution, overall enthalpy change becomes zero.

17 **(d)** $(CH_3)_2CO + CS_2$

Explanation:

Positive deviation occurs when unlike interactions are weaker.

18 **(b) Negative deviation from Raoult's law**

Explanation:

Hydrogen bonding strengthens intermolecular attraction.

19 **(a)** $\Delta H_m = \Delta V_m = 0$

20 **(d) Practically no deviation from Raoult's law**

Explanation:

Benzene and toluene have similar molecular size and intermolecular forces, so they behave nearly ideally.